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SUMMARY

Artificial Intelligence and Data Engineering graduate from Istanbul Technical University with hands-on experience in natural language processing, computer vision, large language models, and end-to-end model development.

WORK EXPERIENCE

ARTIFICIAL INTELLIGENCE INTERN AT NETMERA

Jul 2026 – Present

Developed RAG pipelines and LLM-powered chatbots for retrieving company-specific information and answering customer queries. Built customer journey simulation and predictive modeling solutions to model customer behavior and support data-driven customer engagement.

RESEARCH INTERN AT ÖZYEĞİN UNIVERSITY

Aug 2025 – Oct 2025

Conducted research on speech processing and machine learning for health-related applications; implemented wav2vec/Whisper-based classifiers, built a full dataset preprocessing pipeline, identified and corrected speaker-leakage issues in prior work, and conducted literature analysis and model evaluations.

ARTIFICIAL INTELLIGENCE INTERN AT TURKCELL

Jul 2025 – Aug 2025

Worked on LLM-based intent classification for Turkcell's customer chatbot. Built a RAG pipeline using FAISS, BGE embeddings, and rerankers; optimized prompts with DSPy; evaluated system performance on 9k samples; and collaborated with engineering teams to improve intent taxonomy and retrieval quality.

EDUCATION

Istanbul Technical University

Sep 2021 – Jun 2026

BSc in Artificial Intelligence and Data Engineering

PROJECTS

MONOCULAR DEPTH ESTIMATION & OBSTACLE DETECTION

- Built an end-to-end single-camera obstacle-detection pipeline using Depth Anything V2 and MiDaS, including relative-to-metric depth alignment and distance-thresholded obstacle masking; benchmarked 4 models on the NYU Depth V2 dataset.
- Evaluated depth accuracy, obstacle-detection performance, and inference latency across models, selecting Depth Anything V2 ViT-S as the best accuracy–speed trade-off at 42 FPS on an A100 GPU.
- Fine-tuned Depth Anything V2 ViT-S on 30K RGB-D pairs using hyperparameter search, freezing-strategy ablations, and mixed-precision training, reducing AbsRel by 40% and improving obstacle-detection F1 from 0.48 to 0.66.

SPEECH-BASED HEALTH ASSESSMENT WITH MULTIMODAL DEEP LEARNING

- Designed and implemented an end-to-end multimodal, multi-task deep learning pipeline for 5-class emotion classification, depression detection, and pain detection from speech, fusing acoustic representations from Whisper-base with linguistic representations from BERT using cross-attention.
- Trained and evaluated the system on a combined dataset of 26,382 audio clips/utterances and conducted ablation experiments on different encoder fine-tuning and fusion strategies.
- Achieved a 0.6517 mean macro-F1 on the held-out test set, improving on the baseline by 0.0445.

MULTILINGUAL IDIOM IDENTIFICATION (TURKISH & ITALIAN)

- Developed a BERT–BiLSTM–CRF sequence labeling pipeline for token-level identification of idiomatic multi-word expressions in Turkish and Italian, trained on a multilingual corpus of 68K sentences and 1.48M tokens.
- Applied two-stage fine-tuning with encoder freezing and gradual unfreezing, achieving 0.90 F1 on the test set.

More projects and details: <https://berkekurt.com/#projects>

SKILLS

Python, C++, PyTorch, TensorFlow, Scikit-learn, NLP, Computer Vision, Transformers, XGBoost, CatBoost, Whisper, FAISS, SQL, speech processing, RAG, machine learning, deep learning, Docker, Git